

Model Predictor Testing and Optimization

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Functions Used: modfunctions_notes.R + modelfunctions.R

Model I

Table 1: LN_SPPSF model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19714.49	45	NA	NA	0.1662	0.5329	0.4302	NA	NA	31281.11	31622.16

\begin{table}

\caption{LN_SPPSF model coefficients (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.0550	0.0137	296.4117	0.0000	NA	NA	NA	NA
parametric	arterial	-0.0185	0.0146	-1.2639	0.2063	NA	NA	NA	NA
parametric	single_family1story	0.0433	0.0116	3.7189	0.0002	NA	NA	NA	NA
parametric	half_duplex1story	0.0602	0.0484	1.2446	0.2133	NA	NA	NA	NA
parametric	half_duplex2story	0.0318	0.0464	0.6840	0.4940	NA	NA	NA	NA
parametric	smallest	0.1291	0.0152	8.4868	0.0000	NA	NA	NA	NA
parametric	small	0.0479	0.0111	4.2977	0.0000	NA	NA	NA	NA
parametric	large	-0.0040	0.0126	-0.3181	0.7504	NA	NA	NA	NA
parametric	largest	0.0178	0.0230	0.7726	0.4398	NA	NA	NA	NA
parametric	two_plus_baths	0.1448	0.0274	5.2770	0.0000	NA	NA	NA	NA
parametric	powder_room	0.0659	0.0159	4.1324	0.0000	NA	NA	NA	NA
parametric	good	0.3567	0.0294	12.1144	0.0000	NA	NA	NA	NA
parametric	fair	-0.2076	0.0095	-21.8315	0.0000	NA	NA	NA	NA
parametric	poor	-0.3609	0.0217	-16.6150	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.3365	0.0453	-7.4290	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.4634	0.0284	16.3392	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0380	0.0100	-3.8082	0.0001	NA	NA	NA	NA
parametric	crawl	-0.0138	0.0265	-0.5211	0.6023	NA	NA	NA	NA
parametric	slab	-0.0522	0.0313	-1.6639	0.0962	NA	NA	NA	NA
parametric	fireplace	0.1045	0.0097	10.7497	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0769	0.0150	5.1129	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2946	0.1437	-2.0506	0.0403	NA	NA	NA	NA
parametric	wall	-0.1005	0.0501	-2.0063	0.0448	NA	NA	NA	NA
parametric	central_air	0.0921	0.0108	8.5470	0.0000	NA	NA	NA	NA
smooth	s(sresb_yearbuilt)	NA	NA	55.4700	0.0000	NA	NA	7.6054	9
smooth	s(ln_garagespaces)	NA	NA	39.8021	0.0000	NA	NA	1.9035	3
smooth	s(months)	NA	NA	66.1422	0.0000	NA	NA	7.9988	9

\end{table}

Table 2: LN_SPPSF model fit statistics1

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19712.03	59	NA	NA	0.1665	0.5328	0.4302	NA	NA	31281.91	31658.11

\begin{table}

\caption{LN_SPPSF model coefficients1 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.0658	0.0155	261.9751	0.0000	NA	NA	NA	NA
parametric	arterial	-0.0183	0.0146	-1.2500	0.2113	NA	NA	NA	NA
parametric	single_family1story	0.0419	0.0117	3.5855	0.0003	NA	NA	NA	NA
parametric	half_duplex1story	0.0553	0.0485	1.1412	0.2538	NA	NA	NA	NA
parametric	half_duplex2story	0.0262	0.0466	0.5619	0.5742	NA	NA	NA	NA
parametric	smallest	0.0936	0.0275	3.3975	0.0007	NA	NA	NA	NA
parametric	small	0.0321	0.0149	2.1592	0.0308	NA	NA	NA	NA
parametric	large	0.0110	0.0163	0.6770	0.4984	NA	NA	NA	NA
parametric	largest	0.0432	0.0293	1.4744	0.1404	NA	NA	NA	NA
parametric	two_plus_baths	0.0807	0.0599	1.3471	0.1780	NA	NA	NA	NA
parametric	powder_room	0.0291	0.0311	0.9365	0.3490	NA	NA	NA	NA
parametric	good	0.3561	0.0294	12.0942	0.0000	NA	NA	NA	NA
parametric	fair	-0.2079	0.0095	-21.8597	0.0000	NA	NA	NA	NA
parametric	poor	-0.3612	0.0217	-16.6308	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.3374	0.0453	-7.4495	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.4616	0.0294	15.6931	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0396	0.0100	-3.9500	0.0001	NA	NA	NA	NA
parametric	crawl	-0.0148	0.0265	-0.5598	0.5756	NA	NA	NA	NA
parametric	slab	-0.0525	0.0313	-1.6762	0.0937	NA	NA	NA	NA
parametric	fireplace	0.1061	0.0098	10.8547	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0766	0.0150	5.0912	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2882	0.1437	-2.0061	0.0449	NA	NA	NA	NA
parametric	wall	-0.1032	0.0501	-2.0580	0.0396	NA	NA	NA	NA
parametric	central_air	0.0923	0.0108	8.5638	0.0000	NA	NA	NA	NA
smooth	s(lnsbldsf)	NA	NA	0.3690	0.0405	NA	NA	1.0507	9
smooth	s(totalbath)	NA	NA	0.7473	0.0341	NA	NA	1.4202	5
smooth	s(sresb_yearbuilt)	NA	NA	55.3197	0.0000	NA	NA	7.6079	9
smooth	s(ln_garagespaces)	NA	NA	39.7589	0.0000	NA	NA	1.9003	3
smooth	s(months)	NA	NA	66.2831	0.0000	NA	NA	7.9945	9

\end{table}

Table 3: LN_SPPSF model fit statistics2

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19665.55	101	NA	NA	0.2626	0.5007	0.4053	NA	NA	28893.59	29582.4

\begin{table}

\caption{LN_SPPSF model coefficients2 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.0956	0.0156	263.0623	0.0000	NA	NA	NA	NA
parametric	arterial	0.0058	0.0139	0.4130	0.6796	NA	NA	NA	NA
parametric	single_family1story	0.0430	0.0111	3.8889	0.0001	NA	NA	NA	NA
parametric	half_duplex1story	-0.0123	0.0461	-0.2667	0.7897	NA	NA	NA	NA
parametric	half_duplex2story	-0.0377	0.0442	-0.8531	0.3936	NA	NA	NA	NA
parametric	smallest	0.0678	0.0297	2.2798	0.0226	NA	NA	NA	NA
parametric	small	0.0243	0.0153	1.5840	0.1132	NA	NA	NA	NA
parametric	large	0.0016	0.0169	0.0974	0.9224	NA	NA	NA	NA
parametric	largest	0.0325	0.0303	1.0714	0.2840	NA	NA	NA	NA
parametric	two_plus_baths	0.0449	0.0629	0.7145	0.4749	NA	NA	NA	NA
parametric	powder_room	-0.0117	0.0324	-0.3599	0.7189	NA	NA	NA	NA
parametric	good	0.3055	0.0278	10.9749	0.0000	NA	NA	NA	NA
parametric	fair	-0.1816	0.0090	-20.1715	0.0000	NA	NA	NA	NA
parametric	poor	-0.3223	0.0205	-15.7145	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2860	0.0427	-6.6981	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3871	0.0289	13.3848	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0622	0.0097	-6.4113	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0212	0.0251	-0.8448	0.3983	NA	NA	NA	NA
parametric	slab	-0.0853	0.0297	-2.8755	0.0040	NA	NA	NA	NA
parametric	fireplace	0.0909	0.0095	9.6019	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0713	0.0142	5.0203	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2073	0.1352	-1.5327	0.1254	NA	NA	NA	NA
parametric	wall	-0.1356	0.0474	-2.8613	0.0042	NA	NA	NA	NA
parametric	central_air	0.0722	0.0102	7.0853	0.0000	NA	NA	NA	NA
smooth	s(lnsbldsf)	NA	NA	1.4910	0.0002	NA	NA	1.8406	9
smooth	s(totalbath)	NA	NA	1.2556	0.0043	NA	NA	1.6755	5
smooth	s(sresb_yearbuilt)	NA	NA	17.9070	0.0000	NA	NA	6.0246	9
smooth	s(ln_garagespaces)	NA	NA	55.0108	0.0000	NA	NA	1.9881	3
smooth	s(months)	NA	NA	73.8904	0.0000	NA	NA	7.9439	9
smooth	s(dist_henryford_mi)	NA	NA	38.1347	0.0000	NA	NA	6.8401	7
smooth	s(drive_henryford_min)	NA	NA	37.8368	0.0000	NA	NA	6.3004	7
smooth	s(dist_guardian_mi)	NA	NA	49.8633	0.0000	NA	NA	6.3972	7
smooth	s(drive_guardian_min)	NA	NA	79.3687	0.0000	NA	NA	6.4765	7
smooth	s(dist_dequindre_mi)	NA	NA	72.8422	0.0000	NA	NA	6.6377	7
smooth	s(drive_dequindre_min)	NA	NA	64.7843	0.0000	NA	NA	6.3224	7

\end{table}

Table 4: LN_SPPSF model fit statistics3

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19632.19	143	NA	NA	0.3162	0.4817	0.3914	NA	NA	27442.93	28421.75

\begin{table}

\caption{LN_SPPSF model coefficients3 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.1099	0.0160	256.1586	0.0000	NA	NA	NA	NA
parametric	arterial	0.0191	0.0137	1.3965	0.1626	NA	NA	NA	NA
parametric	single_family1story	0.0319	0.0107	2.9874	0.0028	NA	NA	NA	NA
parametric	half_duplex1story	0.0189	0.0445	0.4249	0.6709	NA	NA	NA	NA
parametric	half_duplex2story	-0.0346	0.0427	-0.8087	0.4187	NA	NA	NA	NA
parametric	smallest	0.0477	0.0310	1.5367	0.1244	NA	NA	NA	NA
parametric	small	0.0129	0.0157	0.8211	0.4116	NA	NA	NA	NA
parametric	large	0.0167	0.0177	0.9446	0.3449	NA	NA	NA	NA
parametric	largest	0.0491	0.0316	1.5547	0.1200	NA	NA	NA	NA
parametric	two_plus_baths	0.0320	0.0674	0.4750	0.6348	NA	NA	NA	NA
parametric	powder_room	-0.0211	0.0343	-0.6159	0.5380	NA	NA	NA	NA
parametric	good	0.2825	0.0269	10.5128	0.0000	NA	NA	NA	NA
parametric	fair	-0.1690	0.0087	-19.4257	0.0000	NA	NA	NA	NA
parametric	poor	-0.3013	0.0198	-15.2363	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2415	0.0412	-5.8643	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3727	0.0286	13.0189	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0656	0.0095	-6.8792	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0902	0.0245	-3.6760	0.0002	NA	NA	NA	NA
parametric	slab	-0.1153	0.0288	-4.0023	0.0001	NA	NA	NA	NA
parametric	fireplace	0.0859	0.0094	9.1522	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0542	0.0137	3.9482	0.0001	NA	NA	NA	NA
parametric	elec_baseboard	-0.2220	0.1304	-1.7025	0.0887	NA	NA	NA	NA
parametric	wall	-0.1420	0.0458	-3.1001	0.0019	NA	NA	NA	NA
parametric	central_air	0.0605	0.0098	6.1508	0.0000	NA	NA	NA	NA
smooth	s(lnsbldsf)	NA	NA	3.5684	0.0000	NA	NA	2.7336	9
smooth	s(totalbath)	NA	NA	1.5596	0.0026	NA	NA	1.9787	5
smooth	s(sresb_yearbuilt)	NA	NA	22.3640	0.0000	NA	NA	6.2004	9
smooth	s(ln_garagespaces)	NA	NA	56.3509	0.0000	NA	NA	1.9993	3
smooth	s(months)	NA	NA	79.8694	0.0000	NA	NA	7.9041	9
smooth	s(dist_henryford_mi)	NA	NA	18.6542	0.0000	NA	NA	6.6544	7
smooth	s(drive_henryford_min)	NA	NA	1.8296	0.0020	NA	NA	3.9028	7
smooth	s(dist_guardian_mi)	NA	NA	17.6858	0.0000	NA	NA	6.7504	7
smooth	s(drive_guardian_min)	NA	NA	12.5354	0.0000	NA	NA	6.0065	7
smooth	s(dist_dequindre_mi)	NA	NA	8.4278	0.0000	NA	NA	4.6211	7
smooth	s(drive_dequindre_min)	NA	NA	18.2958	0.0000	NA	NA	5.2062	7
smooth	s(dist_belleisle_mi)	NA	NA	23.7756	0.0000	NA	NA	6.4898	7
smooth	s(drive_belleisle_min)	NA	NA	10.0949	0.0000	NA	NA	5.8153	7
smooth	s(dist_cayairport_mi)	NA	NA	30.6570	0.0000	NA	NA	6.6962	7
smooth	s(drive_cayairport_min)	NA	NA	13.1582	0.0000	NA	NA	6.6096	7
smooth	s(dist_dtw_mi)	NA	NA	24.6386	0.0000	NA	NA	6.6388	7
smooth	s(drive_dtw_min)	NA	NA	8.6050	0.0000	NA	NA	5.6046	7

\end{table}

Table 5: LN_SPPSF model fit statistics4

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19629.1	146	NA	NA	0.3162	0.4817	0.3914	NA	NA	27445.88	28448.61

\begin{table}

\caption{LN_SPPSF model coefficients4 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.1303	0.0291	142.0005	0.0000	NA	NA	NA	NA
parametric	arterial	0.0189	0.0137	1.3861	0.1657	NA	NA	NA	NA
parametric	single_family1story	0.0318	0.0107	2.9721	0.0030	NA	NA	NA	NA
parametric	half_duplex1story	0.0184	0.0445	0.4124	0.6801	NA	NA	NA	NA
parametric	half_duplex2story	-0.0338	0.0427	-0.7915	0.4287	NA	NA	NA	NA
parametric	smallest	0.0472	0.0310	1.5213	0.1282	NA	NA	NA	NA
parametric	small	0.0128	0.0157	0.8135	0.4159	NA	NA	NA	NA
parametric	large	0.0169	0.0177	0.9556	0.3393	NA	NA	NA	NA
parametric	largest	0.0493	0.0316	1.5593	0.1189	NA	NA	NA	NA
parametric	two_plus_baths	0.0315	0.0674	0.4672	0.6403	NA	NA	NA	NA
parametric	powder_room	-0.0212	0.0343	-0.6173	0.5370	NA	NA	NA	NA
parametric	good	0.2820	0.0269	10.4873	0.0000	NA	NA	NA	NA
parametric	fair	-0.1689	0.0087	-19.4225	0.0000	NA	NA	NA	NA
parametric	poor	-0.3016	0.0198	-15.2472	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2417	0.0412	-5.8681	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3729	0.0286	13.0214	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0660	0.0095	-6.9202	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0897	0.0246	-3.6532	0.0003	NA	NA	NA	NA
parametric	slab	-0.1158	0.0289	-4.0108	0.0001	NA	NA	NA	NA
parametric	fireplace	0.0858	0.0094	9.1329	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0544	0.0137	3.9553	0.0001	NA	NA	NA	NA
parametric	elec_baseboard	-0.2211	0.1304	-1.6956	0.0900	NA	NA	NA	NA
parametric	wall	-0.1419	0.0458	-3.0962	0.0020	NA	NA	NA	NA
parametric	central_air	0.0605	0.0098	6.1425	0.0000	NA	NA	NA	NA
parametric	near_downtown_1mi	0.1730	0.2811	0.6154	0.5383	NA	NA	NA	NA
parametric	near_downtown_5mi	0.0273	0.0441	0.6194	0.5357	NA	NA	NA	NA
parametric	near_downtown_10mi	-0.0283	0.0303	-0.9348	0.3499	NA	NA	NA	NA
smooth	s(lnsbldsf)	NA	NA	3.5972	0.0000	NA	NA	2.7393	9
smooth	s(totalbath)	NA	NA	1.5595	0.0026	NA	NA	1.9774	5
smooth	s(sresb_yearbuilt)	NA	NA	22.3391	0.0000	NA	NA	6.2416	9
smooth	s(ln_garagespaces)	NA	NA	56.4744	0.0000	NA	NA	1.9999	3
smooth	s(months)	NA	NA	79.8571	0.0000	NA	NA	7.9057	9
smooth	s(dist_henryford_mi)	NA	NA	16.7191	0.0000	NA	NA	6.6293	7
smooth	s(drive_henryford_min)	NA	NA	1.9728	0.0012	NA	NA	3.9898	7
smooth	s(dist_guardian_mi)	NA	NA	17.8148	0.0000	NA	NA	6.7466	7
smooth	s(drive_guardian_min)	NA	NA	12.3835	0.0000	NA	NA	6.0042	7
smooth	s(dist_dequindre_mi)	NA	NA	8.5449	0.0000	NA	NA	4.6225	7
smooth	s(drive_dequindre_min)	NA	NA	17.8568	0.0000	NA	NA	5.1890	7
smooth	s(dist_belleisle_mi)	NA	NA	23.9028	0.0000	NA	NA	6.4802	7
smooth	s(drive_belleisle_min)	NA	NA	10.0372	0.0000	NA	NA	5.8216	7
smooth	s(dist_cayairport_mi)	NA	NA	30.7526	0.0000	NA	NA	6.6950	7
smooth	s(drive_cayairport_min)	NA	NA	12.7096	0.0000	NA	NA	6.6081	7
smooth	s(dist_dtw_mi)	NA	NA	24.6652	0.0000	NA	NA	6.6417	7
smooth	s(drive_dtw_min)	NA	NA	8.5910	0.0000	NA	NA	5.6111	7

\end{table}

Table 6: LN_SPPSF model fit statistics5

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19612.41	188	NA	NA	0.326	0.478	0.3884	NA	NA	27178.37	28323.87

\begin{table}

\caption{LN_SPPSF model coefficients5 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	4.1601	0.0312	133.1629	0.0000	NA	NA	NA	NA
parametric	arterial	0.0104	0.0137	0.7578	0.4486	NA	NA	NA	NA
parametric	single_family1story	0.0297	0.0106	2.7948	0.0052	NA	NA	NA	NA
parametric	half_duplex1story	0.0035	0.0443	0.0801	0.9362	NA	NA	NA	NA
parametric	half_duplex2story	-0.0476	0.0425	-1.1216	0.2620	NA	NA	NA	NA
parametric	smallest	0.0401	0.0314	1.2774	0.2015	NA	NA	NA	NA
parametric	small	0.0104	0.0159	0.6565	0.5115	NA	NA	NA	NA
parametric	large	0.0197	0.0180	1.0965	0.2729	NA	NA	NA	NA
parametric	largest	0.0548	0.0320	1.7141	0.0865	NA	NA	NA	NA
parametric	two_plus_baths	0.0389	0.0659	0.5901	0.5551	NA	NA	NA	NA
parametric	powder_room	-0.0193	0.0336	-0.5747	0.5655	NA	NA	NA	NA
parametric	good	0.2757	0.0267	10.3120	0.0000	NA	NA	NA	NA
parametric	fair	-0.1661	0.0087	-19.2044	0.0000	NA	NA	NA	NA
parametric	poor	-0.2939	0.0197	-14.9510	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2411	0.0410	-5.8872	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3784	0.0286	13.2179	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0665	0.0096	-6.9552	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0893	0.0245	-3.6528	0.0003	NA	NA	NA	NA
parametric	slab	-0.1155	0.0287	-4.0282	0.0001	NA	NA	NA	NA
parametric	fireplace	0.0801	0.0093	8.5651	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0475	0.0136	3.4771	0.0005	NA	NA	NA	NA
parametric	elec_baseboard	-0.2187	0.1295	-1.6886	0.0913	NA	NA	NA	NA
parametric	wall	-0.1460	0.0456	-3.2055	0.0014	NA	NA	NA	NA
parametric	central_air	0.0573	0.0098	5.8578	0.0000	NA	NA	NA	NA
parametric	near_downtown_1mi	-0.3384	0.3034	-1.1155	0.2646	NA	NA	NA	NA
parametric	near_downtown_5mi	0.0185	0.0446	0.4140	0.6789	NA	NA	NA	NA
parametric	near_downtown_10mi	-0.0595	0.0338	-1.7627	0.0780	NA	NA	NA	NA
smooth	s(lnsbldsf)	NA	NA	4.6017	0.0000	NA	NA	3.0365	9
smooth	s(totalbath)	NA	NA	1.3613	0.0042	NA	NA	1.8459	5
smooth	s(sresb_yearbuilt)	NA	NA	21.1001	0.0000	NA	NA	5.5195	9
smooth	s(ln_garagespaces)	NA	NA	55.3460	0.0000	NA	NA	1.9993	3
smooth	s(months)	NA	NA	81.3092	0.0000	NA	NA	7.9258	9
smooth	s(dist_henryford_mi)	NA	NA	20.5541	0.0000	NA	NA	6.7764	7
smooth	s(drive_henryford_min)	NA	NA	2.7336	0.0000	NA	NA	2.3321	7
smooth	s(dist_guardian_mi)	NA	NA	5.7734	0.0000	NA	NA	4.4990	7
smooth	s(drive_guardian_min)	NA	NA	10.2835	0.0000	NA	NA	6.1564	7
smooth	s(dist_dequindre_mi)	NA	NA	0.0000	0.4581	NA	NA	0.0001	7
smooth	s(drive_dequindre_min)	NA	NA	15.9207	0.0000	NA	NA	4.8283	7
smooth	s(dist_belleisle_mi)	NA	NA	16.7982	0.0000	NA	NA	6.2979	7
smooth	s(drive_belleisle_min)	NA	NA	10.5162	0.0000	NA	NA	6.0434	7
smooth	s(dist_cayairport_mi)	NA	NA	26.5935	0.0000	NA	NA	6.5930	7
smooth	s(drive_cayairport_min)	NA	NA	9.0373	0.0000	NA	NA	6.2877	7
smooth	s(dist_dtw_mi)	NA	NA	16.8937	0.0000	NA	NA	5.9580	7
smooth	s(drive_dtw_min)	NA	NA	5.8216	0.0000	NA	NA	4.7334	7
smooth	s(dist_palmer_mi)	NA	NA	10.1392	0.0000	NA	NA	6.3143	7
smooth	s(drive_palmer_min)	NA	NA	11.3063	0.0000	NA	NA	6.5117	7
smooth	s(dist_roosevelt_mi)	NA	NA	19.4812	0.0000	NA	NA	6.6067	7
smooth	s(drive_roosevelt_min)	NA	NA	6.8244	0.0000	NA	NA	5.7641	7
smooth	s(dist_dmc_mi)	NA	NA	1.5839	0.0006	NA	NA	2.5509	7
smooth	s(drive_dmc_min)	NA	NA	0.0007	0.4889	NA	NA	0.0109	7

\end{table}

Model II

Table 7: LN_PRICE model fit statistics

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19703.32	63	NA	NA	0.3778	0.5289	0.4266	NA	NA	31013.8	31458.33

\begin{table}

\caption{LN_PRICE model coefficients (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	11.1388	0.0166	669.8558	0.0000	NA	NA	NA	NA
parametric	arterial	-0.0111	0.0146	-0.7638	0.4450	NA	NA	NA	NA
parametric	single_family1story	0.0336	0.0117	2.8728	0.0041	NA	NA	NA	NA
parametric	half_duplex1story	-0.1539	0.0498	-3.0892	0.0020	NA	NA	NA	NA
parametric	half_duplex2story	-0.1667	0.0481	-3.4653	0.0005	NA	NA	NA	NA
parametric	smallest	0.0771	0.0363	2.1256	0.0335	NA	NA	NA	NA
parametric	small	0.0195	0.0184	1.0615	0.2885	NA	NA	NA	NA
parametric	large	0.0006	0.0212	0.0273	0.9782	NA	NA	NA	NA
parametric	largest	0.0106	0.0370	0.2866	0.7744	NA	NA	NA	NA
parametric	two_plus_baths	0.1642	0.0409	4.0115	0.0001	NA	NA	NA	NA
parametric	powder_room	0.0593	0.0196	3.0266	0.0025	NA	NA	NA	NA
parametric	good	0.3564	0.0292	12.1861	0.0000	NA	NA	NA	NA
parametric	fair	-0.2015	0.0095	-21.3058	0.0000	NA	NA	NA	NA
parametric	poor	-0.3526	0.0216	-16.3417	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.3304	0.0450	-7.3453	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.4474	0.0296	15.1101	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0433	0.0100	-4.3183	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0173	0.0265	-0.6542	0.5130	NA	NA	NA	NA
parametric	slab	-0.0572	0.0313	-1.8291	0.0674	NA	NA	NA	NA
parametric	fireplace	0.1012	0.0098	10.3194	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0730	0.0150	4.8703	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2987	0.1430	-2.0888	0.0367	NA	NA	NA	NA
parametric	wall	-0.1131	0.0498	-2.2695	0.0232	NA	NA	NA	NA
parametric	central_air	0.0895	0.0107	8.3663	0.0000	NA	NA	NA	NA
smooth	s(ln_landratio)	NA	NA	30.8565	0.0000	NA	NA	6.7570	9
smooth	s(ln_base_bldgsize_ratio)	NA	NA	29.1658	0.0000	NA	NA	4.4864	9
smooth	s(sresb_yearbuilt)	NA	NA	35.6394	0.0000	NA	NA	7.6205	9
smooth	s(ln_garagespaces)	NA	NA	25.2660	0.0000	NA	NA	1.8004	3
smooth	s(months)	NA	NA	68.7955	0.0000	NA	NA	8.0137	9

\end{table}

Table 8: LN_PRICE model fit statistics1

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19701.61	77	NA	NA	0.378	0.5288	0.4265	NA	NA	31012.33	31484.88

\begin{table}

\caption{LN_PRICE model coefficients1 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	11.1492	0.0184	604.7293	0.0000	NA	NA	NA	NA
parametric	arterial	-0.0108	0.0146	-0.7383	0.4603	NA	NA	NA	NA
parametric	single_family1story	0.0335	0.0117	2.8702	0.0041	NA	NA	NA	NA
parametric	half_duplex1story	0.0366	0.0659	0.5559	0.5783	NA	NA	NA	NA
parametric	half_duplex2story	0.0161	0.0646	0.2501	0.8025	NA	NA	NA	NA
parametric	smallest	0.0801	0.0368	2.1752	0.0296	NA	NA	NA	NA
parametric	small	0.0152	0.0188	0.8104	0.4177	NA	NA	NA	NA
parametric	large	0.0048	0.0218	0.2223	0.8241	NA	NA	NA	NA
parametric	largest	0.0165	0.0380	0.4341	0.6642	NA	NA	NA	NA
parametric	two_plus_baths	0.0530	0.0734	0.7226	0.4700	NA	NA	NA	NA
parametric	powder_room	0.0000	0.0368	0.0000	1.0000	NA	NA	NA	NA
parametric	good	0.3564	0.0293	12.1837	0.0000	NA	NA	NA	NA
parametric	fair	-0.2016	0.0095	-21.3209	0.0000	NA	NA	NA	NA
parametric	poor	-0.3525	0.0216	-16.3379	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.3315	0.0450	-7.3689	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.4438	0.0298	14.9071	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0435	0.0100	-4.3347	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0162	0.0265	-0.6140	0.5392	NA	NA	NA	NA
parametric	slab	-0.0554	0.0313	-1.7726	0.0763	NA	NA	NA	NA
parametric	fireplace	0.1016	0.0098	10.3593	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0719	0.0150	4.7964	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2912	0.1430	-2.0363	0.0417	NA	NA	NA	NA
parametric	wall	-0.1130	0.0498	-2.2669	0.0234	NA	NA	NA	NA
parametric	central_air	0.0895	0.0107	8.3632	0.0000	NA	NA	NA	NA
smooth	s(ln_landratio)	NA	NA	30.8314	0.0000	NA	NA	6.7590	9
smooth	s(ln_base_bldgsize_ratio)	NA	NA	0.0259	0.0192	NA	NA	0.2453	9
smooth	s(lnsbldsf)	NA	NA	2.5734	0.0000	NA	NA	4.3619	9
smooth	s(totalbath)	NA	NA	1.0194	0.0115	NA	NA	1.5864	5
smooth	s(sresb_yearbuilt)	NA	NA	35.3527	0.0000	NA	NA	7.6266	9
smooth	s(ln_garagespaces)	NA	NA	25.0855	0.0000	NA	NA	1.7987	3
smooth	s(months)	NA	NA	68.7496	0.0000	NA	NA	8.0155	9

\end{table}

Table 9: LN_PRICE model fit statistics2

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19657.5	105	NA	NA	0.4446	0.4993	0.4042	NA	NA	28805.07	29566.54

\begin{table}

\caption{LN_PRICE model coefficients2 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	11.1533	0.0153	729.3975	0.0000	NA	NA	NA	NA
parametric	arterial	0.0097	0.0139	0.6945	0.4874	NA	NA	NA	NA
parametric	single_family1story	0.0390	0.0111	3.5260	0.0004	NA	NA	NA	NA
parametric	half_duplex1story	-0.2015	0.0474	-4.2508	0.0000	NA	NA	NA	NA
parametric	half_duplex2story	-0.2197	0.0455	-4.8253	0.0000	NA	NA	NA	NA
parametric	smallest	0.0654	0.0332	1.9715	0.0487	NA	NA	NA	NA
parametric	small	0.0203	0.0168	1.2052	0.2282	NA	NA	NA	NA
parametric	large	-0.0016	0.0191	-0.0860	0.9315	NA	NA	NA	NA
parametric	largest	0.0198	0.0336	0.5902	0.5550	NA	NA	NA	NA
parametric	two_plus_baths	0.1425	0.0373	3.8187	0.0001	NA	NA	NA	NA
parametric	powder_room	0.0339	0.0180	1.8890	0.0589	NA	NA	NA	NA
parametric	good	0.3039	0.0278	10.9425	0.0000	NA	NA	NA	NA
parametric	fair	-0.1796	0.0090	-19.9876	0.0000	NA	NA	NA	NA
parametric	poor	-0.3200	0.0205	-15.6386	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2859	0.0426	-6.7128	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3742	0.0291	12.8788	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0625	0.0097	-6.4252	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0231	0.0252	-0.9154	0.3600	NA	NA	NA	NA
parametric	slab	-0.0892	0.0297	-3.0012	0.0027	NA	NA	NA	NA
parametric	fireplace	0.0875	0.0095	9.1846	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0707	0.0142	4.9708	0.0000	NA	NA	NA	NA
parametric	elec_baseboard	-0.2185	0.1352	-1.6158	0.1062	NA	NA	NA	NA
parametric	wall	-0.1340	0.0473	-2.8315	0.0046	NA	NA	NA	NA
parametric	central_air	0.0705	0.0102	6.9321	0.0000	NA	NA	NA	NA
smooth	s(ln_landratio)	NA	NA	11.4292	0.0000	NA	NA	7.4873	9
smooth	s(ln_base_bldgsize_ratio)	NA	NA	27.8512	0.0000	NA	NA	3.7155	9
smooth	s(sresb_yearbuilt)	NA	NA	16.6747	0.0000	NA	NA	6.4423	9
smooth	s(ln_garagespaces)	NA	NA	47.5566	0.0000	NA	NA	1.9481	3
smooth	s(months)	NA	NA	75.0291	0.0000	NA	NA	7.9640	9
smooth	s(dist_henryford_mi)	NA	NA	39.9436	0.0000	NA	NA	6.8515	7
smooth	s(drive_henryford_min)	NA	NA	37.5965	0.0000	NA	NA	6.3021	7
smooth	s(dist_guardian_mi)	NA	NA	49.2735	0.0000	NA	NA	6.3649	7
smooth	s(drive_guardian_min)	NA	NA	77.2056	0.0000	NA	NA	6.4782	7
smooth	s(dist_dequindre_mi)	NA	NA	73.3090	0.0000	NA	NA	6.6824	7
smooth	s(drive_dequindre_min)	NA	NA	63.9223	0.0000	NA	NA	6.2612	7

\end{table}

Table 10: LN_PRICE model fit statistics3

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19627.36	147	NA	NA	0.4847	0.4806	0.3904	NA	NA	27358.78	28373.3

\begin{table}

\caption{LN_PRICE model coefficients3 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	11.1659	0.0149	749.1893	0.0000	NA	NA	NA	NA
parametric	arterial	0.0206	0.0136	1.5090	0.1313	NA	NA	NA	NA
parametric	single_family1story	0.0283	0.0107	2.6484	0.0081	NA	NA	NA	NA
parametric	half_duplex1story	-0.1577	0.0458	-3.4416	0.0006	NA	NA	NA	NA
parametric	half_duplex2story	-0.2032	0.0440	-4.6188	0.0000	NA	NA	NA	NA
parametric	smallest	0.0508	0.0323	1.5714	0.1161	NA	NA	NA	NA
parametric	small	0.0106	0.0164	0.6490	0.5164	NA	NA	NA	NA
parametric	large	0.0152	0.0187	0.8158	0.4146	NA	NA	NA	NA
parametric	largest	0.0362	0.0328	1.1046	0.2694	NA	NA	NA	NA
parametric	two_plus_baths	0.1441	0.0364	3.9547	0.0001	NA	NA	NA	NA
parametric	powder_room	0.0330	0.0175	1.8914	0.0586	NA	NA	NA	NA
parametric	good	0.2826	0.0268	10.5415	0.0000	NA	NA	NA	NA
parametric	fair	-0.1680	0.0087	-19.3459	0.0000	NA	NA	NA	NA
parametric	poor	-0.2996	0.0197	-15.1821	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2398	0.0411	-5.8358	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3579	0.0286	12.4951	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0653	0.0095	-6.8529	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0840	0.0245	-3.4211	0.0006	NA	NA	NA	NA
parametric	slab	-0.1165	0.0288	-4.0419	0.0001	NA	NA	NA	NA
parametric	fireplace	0.0791	0.0094	8.4042	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0486	0.0138	3.5243	0.0004	NA	NA	NA	NA
parametric	elec_baseboard	-0.2324	0.1303	-1.7832	0.0746	NA	NA	NA	NA
parametric	wall	-0.1468	0.0457	-3.2095	0.0013	NA	NA	NA	NA
parametric	central_air	0.0583	0.0098	5.9302	0.0000	NA	NA	NA	NA
smooth	s(ln_landratio)	NA	NA	9.7774	0.0000	NA	NA	4.9598	9
smooth	s(ln_base_bldgsize_ratio)	NA	NA	23.5079	0.0000	NA	NA	3.8883	9
smooth	s(sresb_yearbuilt)	NA	NA	19.0027	0.0000	NA	NA	6.5856	9
smooth	s(ln_garagespaces)	NA	NA	46.5743	0.0000	NA	NA	1.9457	3
smooth	s(months)	NA	NA	81.3792	0.0000	NA	NA	7.9072	9
smooth	s(dist_henryford_mi)	NA	NA	18.9081	0.0000	NA	NA	6.6713	7
smooth	s(drive_henryford_min)	NA	NA	2.2646	0.0005	NA	NA	4.1868	7
smooth	s(dist_guardian_mi)	NA	NA	18.3405	0.0000	NA	NA	6.7660	7
smooth	s(drive_guardian_min)	NA	NA	11.9208	0.0000	NA	NA	5.9369	7
smooth	s(dist_dequindre_mi)	NA	NA	9.3755	0.0000	NA	NA	4.8997	7
smooth	s(drive_dequindre_min)	NA	NA	18.5348	0.0000	NA	NA	5.2473	7
smooth	s(dist_belleisle_mi)	NA	NA	24.0753	0.0000	NA	NA	6.5172	7
smooth	s(drive_belleisle_min)	NA	NA	9.4396	0.0000	NA	NA	5.6735	7
smooth	s(dist_cayairport_mi)	NA	NA	33.1496	0.0000	NA	NA	6.7333	7
smooth	s(drive_cayairport_min)	NA	NA	13.0919	0.0000	NA	NA	6.6137	7
smooth	s(dist_dtw_mi)	NA	NA	25.7997	0.0000	NA	NA	6.6329	7
smooth	s(drive_dtw_min)	NA	NA	7.7236	0.0000	NA	NA	5.4710	7

\end{table}

Table 11: LN_PRICE model fit statistics4

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19624.29	150	NA	NA	0.4847	0.4806	0.3904	NA	NA	27362.35	28400.76

\begin{table}

\caption{LN_PRICE model coefficients4 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	11.1801	0.0285	392.4224	0.0000	NA	NA	NA	NA
parametric	arterial	0.0204	0.0137	1.4924	0.1356	NA	NA	NA	NA
parametric	single_family1story	0.0281	0.0107	2.6291	0.0086	NA	NA	NA	NA
parametric	half_duplex1story	-0.1584	0.0459	-3.4541	0.0006	NA	NA	NA	NA
parametric	half_duplex2story	-0.2024	0.0440	-4.5976	0.0000	NA	NA	NA	NA
parametric	smallest	0.0508	0.0323	1.5706	0.1163	NA	NA	NA	NA
parametric	small	0.0106	0.0164	0.6498	0.5158	NA	NA	NA	NA
parametric	large	0.0153	0.0187	0.8183	0.4132	NA	NA	NA	NA
parametric	largest	0.0363	0.0328	1.1088	0.2675	NA	NA	NA	NA
parametric	two_plus_baths	0.1441	0.0365	3.9520	0.0001	NA	NA	NA	NA
parametric	powder_room	0.0332	0.0175	1.9000	0.0574	NA	NA	NA	NA
parametric	good	0.2819	0.0268	10.5082	0.0000	NA	NA	NA	NA
parametric	fair	-0.1679	0.0087	-19.3411	0.0000	NA	NA	NA	NA
parametric	poor	-0.3000	0.0197	-15.1944	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2401	0.0411	-5.8423	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3584	0.0287	12.5042	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0656	0.0095	-6.8800	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0835	0.0246	-3.4004	0.0007	NA	NA	NA	NA
parametric	slab	-0.1172	0.0289	-4.0578	0.0000	NA	NA	NA	NA
parametric	fireplace	0.0790	0.0094	8.3889	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0486	0.0138	3.5282	0.0004	NA	NA	NA	NA
parametric	elec_baseboard	-0.2317	0.1303	-1.7777	0.0755	NA	NA	NA	NA
parametric	wall	-0.1463	0.0457	-3.1983	0.0014	NA	NA	NA	NA
parametric	central_air	0.0583	0.0098	5.9270	0.0000	NA	NA	NA	NA
parametric	near_downtown_1mi	0.2222	0.2811	0.7905	0.4293	NA	NA	NA	NA
parametric	near_downtown_5mi	0.0305	0.0441	0.6927	0.4885	NA	NA	NA	NA
parametric	near_downtown_10mi	-0.0210	0.0304	-0.6903	0.4900	NA	NA	NA	NA
smooth	s(ln_landratio)	NA	NA	9.7629	0.0000	NA	NA	4.9602	9
smooth	s(ln_base_bldgsize_ratio)	NA	NA	23.4249	0.0000	NA	NA	3.8926	9
smooth	s(sresb_yearbuilt)	NA	NA	18.9402	0.0000	NA	NA	6.6271	9
smooth	s(ln_garagespaces)	NA	NA	46.6879	0.0000	NA	NA	1.9459	3
smooth	s(months)	NA	NA	81.3759	0.0000	NA	NA	7.9080	9
smooth	s(dist_henryford_mi)	NA	NA	17.0523	0.0000	NA	NA	6.6507	7
smooth	s(drive_henryford_min)	NA	NA	2.4154	0.0003	NA	NA	4.2592	7
smooth	s(dist_guardian_mi)	NA	NA	18.5118	0.0000	NA	NA	6.7627	7
smooth	s(drive_guardian_min)	NA	NA	11.7909	0.0000	NA	NA	5.9388	7
smooth	s(dist_dequindre_mi)	NA	NA	9.3930	0.0000	NA	NA	4.8734	7
smooth	s(drive_dequindre_min)	NA	NA	18.0647	0.0000	NA	NA	5.2259	7
smooth	s(dist_belleisle_mi)	NA	NA	24.1886	0.0000	NA	NA	6.5050	7
smooth	s(drive_belleisle_min)	NA	NA	9.4011	0.0000	NA	NA	5.6898	7
smooth	s(dist_cayairport_mi)	NA	NA	33.1390	0.0000	NA	NA	6.7281	7
smooth	s(drive_cayairport_min)	NA	NA	12.7636	0.0000	NA	NA	6.6163	7
smooth	s(dist_dtw_mi)	NA	NA	25.6671	0.0000	NA	NA	6.6341	7
smooth	s(drive_dtw_min)	NA	NA	7.7658	0.0000	NA	NA	5.4889	7

\end{table}

Table 12: LN_PRICE model fit statistics5

n	df_res	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19748	19607.99	192	NA	NA	0.4923	0.4768	0.3874	NA	NA	27091.02	28277.76

\begin{table}

\caption{LN_PRICE model coefficients5 (95% CI)}

Type	Term	Estimate	Std. Error	t value	p value	CI low	CI high	edf	ref df
parametric	(Intercept)	11.2091	0.0306	365.7889	0.0000	NA	NA	NA	NA
parametric	arterial	0.0118	0.0137	0.8647	0.3872	NA	NA	NA	NA
parametric	single_family1story	0.0259	0.0106	2.4418	0.0146	NA	NA	NA	NA
parametric	half_duplex1story	-0.1682	0.0456	-3.6919	0.0002	NA	NA	NA	NA
parametric	half_duplex2story	-0.2104	0.0437	-4.8096	0.0000	NA	NA	NA	NA
parametric	smallest	0.0425	0.0323	1.3170	0.1879	NA	NA	NA	NA
parametric	small	0.0084	0.0163	0.5122	0.6085	NA	NA	NA	NA
parametric	large	0.0174	0.0186	0.9338	0.3504	NA	NA	NA	NA
parametric	largest	0.0412	0.0327	1.2584	0.2082	NA	NA	NA	NA
parametric	two_plus_baths	0.1425	0.0364	3.9163	0.0001	NA	NA	NA	NA
parametric	powder_room	0.0323	0.0174	1.8553	0.0636	NA	NA	NA	NA
parametric	good	0.2757	0.0267	10.3370	0.0000	NA	NA	NA	NA
parametric	fair	-0.1651	0.0086	-19.1199	0.0000	NA	NA	NA	NA
parametric	poor	-0.2920	0.0196	-14.8899	0.0000	NA	NA	NA	NA
parametric	very_poor_unsound	-0.2396	0.0409	-5.8635	0.0000	NA	NA	NA	NA
parametric	qabove_avg	0.3632	0.0286	12.6831	0.0000	NA	NA	NA	NA
parametric	qbelow_avg	-0.0656	0.0096	-6.8673	0.0000	NA	NA	NA	NA
parametric	crawl	-0.0806	0.0245	-3.2935	0.0010	NA	NA	NA	NA
parametric	slab	-0.1150	0.0287	-4.0126	0.0001	NA	NA	NA	NA
parametric	fireplace	0.0728	0.0094	7.7644	0.0000	NA	NA	NA	NA
parametric	hot_water	0.0413	0.0137	3.0163	0.0026	NA	NA	NA	NA
parametric	elec_baseboard	-0.2242	0.1294	-1.7320	0.0833	NA	NA	NA	NA
parametric	wall	-0.1510	0.0455	-3.3213	0.0009	NA	NA	NA	NA
parametric	central_air	0.0553	0.0098	5.6596	0.0000	NA	NA	NA	NA
parametric	near_downtown_1mi	-0.2859	0.3034	-0.9422	0.3461	NA	NA	NA	NA
parametric	near_downtown_5mi	0.0241	0.0446	0.5401	0.5892	NA	NA	NA	NA
parametric	near_downtown_10mi	-0.0500	0.0338	-1.4799	0.1389	NA	NA	NA	NA
smooth	s(ln_landratio)	NA	NA	10.9189	0.0000	NA	NA	4.9313	9
smooth	s(ln_base_bldgsize_ratio)	NA	NA	21.3068	0.0000	NA	NA	3.9815	9
smooth	s(sresb_yearbuilt)	NA	NA	17.2449	0.0000	NA	NA	5.7587	9
smooth	s(ln_garagespaces)	NA	NA	45.2283	0.0000	NA	NA	1.9377	3
smooth	s(months)	NA	NA	83.0745	0.0000	NA	NA	7.9280	9
smooth	s(dist_henryford_mi)	NA	NA	21.1113	0.0000	NA	NA	6.7752	7
smooth	s(drive_henryford_min)	NA	NA	3.0115	0.0000	NA	NA	2.5136	7
smooth	s(dist_guardian_mi)	NA	NA	7.2199	0.0000	NA	NA	4.6815	7
smooth	s(drive_guardian_min)	NA	NA	9.9789	0.0000	NA	NA	6.1295	7
smooth	s(dist_dequindre_mi)	NA	NA	0.0001	0.3552	NA	NA	0.0005	7
smooth	s(drive_dequindre_min)	NA	NA	16.1707	0.0000	NA	NA	4.8729	7
smooth	s(dist_belleisle_mi)	NA	NA	16.9682	0.0000	NA	NA	6.3063	7
smooth	s(drive_belleisle_min)	NA	NA	10.2662	0.0000	NA	NA	5.9847	7
smooth	s(dist_cayairport_mi)	NA	NA	28.0502	0.0000	NA	NA	6.6174	7
smooth	s(drive_cayairport_min)	NA	NA	9.1244	0.0000	NA	NA	6.3117	7
smooth	s(dist_dtw_mi)	NA	NA	16.7226	0.0000	NA	NA	5.9809	7
smooth	s(drive_dtw_min)	NA	NA	5.2298	0.0000	NA	NA	4.6069	7
smooth	s(dist_palmer_mi)	NA	NA	10.3852	0.0000	NA	NA	6.2636	7
smooth	s(drive_palmer_min)	NA	NA	12.0016	0.0000	NA	NA	6.5389	7
smooth	s(dist_roosevelt_mi)	NA	NA	20.0021	0.0000	NA	NA	6.6123	7
smooth	s(drive_roosevelt_min)	NA	NA	7.1181	0.0000	NA	NA	5.8244	7
smooth	s(dist_dmc_mi)	NA	NA	1.4330	0.0012	NA	NA	2.4393	7
smooth	s(drive_dmc_min)	NA	NA	0.0006	0.5828	NA	NA	0.0128	7

\end{table}